Infosys - Coding 10

Deep within the mystical realm of Sylvaria, the sacred Elden Tree stands tall, its ancient roots binding the fate of the land.

You are given a tree with N nodes, where each node represents a guardian spirit. The tree is rooted at node 1.

Each node i (except the root) has:

1. A parent node P[i], which tells us which node it is directly connected to.
2. A weight W[i], which represents the strength of the connection (Lifeline) between i and its parent P[i].

For every pair of nodes (u,v), determine:

1. The strongest Lifeline (i.e., the maximum weight edge) in the shortest path between u and v, and sum it across all pairs — call it MaxLifelineSum.
2. The weakest Lifeline (i.e., the minimum weight edge) in the shortest path between u and v, and sum it across all pairs — call it MinLifelineSum.

Return these two sums as an array arr containing [MaxLifelineSum, MinLifelineSum].

Input Format

The first line contains an integer, N, denoting the number of elements in P.

Each line i of the N subsequent lines (where 1 ≤ i ≤ N) contains an integer describing "P[i]". First is always = -1

Each line i of the N subsequent lines (where 1 ≤ i ≤ N) contains an integer describing "W[i]". First is always = -1

Constraints

1 <= N <= 10^5

1 <= P[i] <= 10^5

1 <= W[i] <= 10^5

Output Format

The output contain integer values on both the lines.

Sample Input 0

```
2
-1
1
```

```
-1
5000
```

Sample Output 0

```
5000
5000
```

Explanation 0

N=2, P=[-1 1], W=[-1 5000] (1) | (2) [5000] Pairs & Calculations The only pair is (1,2), connected by a single edge 5000. Strongest Lifeline in the path = 5000 Weakest Lifeline in the path = 5000

Sample Input 1

```
3
-1
1
2
-1
1
2
```

Sample Output 1

```
5
4
```

Explanation 1

N=3, P=[-1 1 2], W=[-1 1 2] (1) | (2) [1] | (3) [2] Pairs & Calculations (1,2): Path → [1 → 2], Max = 1, Min = 1 (2,3): Path → [2 → 3], Max = 2, Min = 2 (1,3): Path → [1 → 2 → 3], Max = 2, Min = 1 Strongest Lifeline = 1 + 2 + 2 = 5 Weakest Lifeline = 1 + 1 + 2 = 4